Certificate Generator

Web-Based Certificate Generator using Flutter Web

1.0 Introduction

1.1 Project Background

- Certificates are widely used for recognition, participation, and completion
- Manual preparation involves design, data entry, and document production
- Existing tools require combining multiple applications for design, data, and generation
- **Proposed solution:** A web-based certificate generator using Flutter Web
  - Layered certificate design + template reuse
  - Manual entry and CSV import for recipient data
  - Batch generation into a single PDF
  - User accounts for persistent storage

1.2 Problem Statement

- **Repetitive manual effort:** Each certificate is edited individually → errors in names, dates, and positioning
- **Poor reuse and management:** Templates and recipient records scattered across files/apps
- **Scaling issues:** Large groups need reliable data-to-field association
- **Need:** Centralized, automated system with consistent PDF output

1.3 Project Objectives

1. Develop a **web-based certificate generator** with layered design elements and reusable templates
2. Implement a **centralized management system** for templates and recipient data via user accounts
3. Develop **individual and batch generation** supporting manual entry and CSV imports ⟶ consistent PDF output

2.0 Literature Review

- Reviews existing approaches, technologies, and systems for certificate generation
- Focus areas:
  - Template customization
  - Recipient data management
  - Automated PDF generation
- Examines existing applications and development technologies
- Identifies relevant features, limitations, and opportunities
- **Note:** Research based on articles, papers, and competitor analysis (e.g., alternativeTo.com batch certificate generators)

2.1 Feasibility Studies

- **(Section placeholder — to be completed)**
- Key feasibility dimensions to cover:
  - Technical feasibility
  - Economic feasibility
  - Operational feasibility
  - Schedule feasibility

3.0 Project Methodology

- **Selected SDLC Model:** Throwaway Prototyping
- Rationale: Test technical features before finalizing design
- Reduces risks in:
  - Layered certificate editing
  - CSV data processing
  - PDF generation

3.1.1 SDLC Phases

1. **Planning & Requirements Analysis**
   - Identify requirements: template customization, authentication, data management, generation
2. **Prototype Design & Construction**
   - Test text positioning, layered elements, CSV imports, PDF generation
3. **Prototype Evaluation**
   - Identify technical issues; refine final design
4. **Final System Design & Implementation**
   - Build with Flutter Web + self-hosted backend
   - Discard prototypes; apply lessons learned

3.1 Principal of Operation

- Users log in to access saved templates and recipient data
- **Design:** Create from scratch with layered elements or upload existing background
- **Customize:** Position and format text elements (names, titles, dates)
- **Save:** Store templates for future use
- **Data:** Manual entry or CSV import
- **Generate:** Apply data to template ⟶ individual or batch
- **Output:** Combined PDF for review, printing, and distribution

3.1.2 Use Case Diagram

- **(Diagram placeholder — insert use case diagram)**
- Key actors:
  - Student / User
  - Admin
  - Backend Service
- Main use cases:
  - UC01: User Login
  - Create/Manage Templates
  - Manage Recipient Data
  - CSV Import
  - Generate Individual Certificate

- Generate Batch Certificates
- Export/Print PDF
- Admin Account Management
- Backend Service Operations

3.1.3 Use Case Specification (Summary)

UC01: User Login

- Actor: Student / User
- Precondition: Registered account
- Flow: Open app → enter credentials → validate → authenticate → dashboard
- Postcondition: Access to saved templates and recipient data

UC: CSV Import

- Actor: Student / User
- Precondition: Logged in; CSV file available

- Flow: Upload → read headers → map fields → validate → preview → confirm → save
- Postcondition: Valid records imported for generation

3.1.3 Use Case Specification (Cont.)

UC: Generate Individual Certificate

- Select template → select/enter recipient → validate → insert details → preview → export PDF

UC: Generate Batch Certificates

- Select template → select dataset/CSV → validate → preview → confirm → apply to each record → combine into one PDF → export

UC: Export / Print PDF

- Review pages → select pages (if supported) → export or print

UC: Admin Account Management

- Search/select account → view info → deactivate/delete → confirm → apply change

3.2.1 Software Specification

Software	Version	Justification
Chromium Browser	Latest stable	Runs Flutter Web app; preview, uploads, PDF downloads
Flutter SDK	Stable release	Develop UI and application logic
Dart SDK	Bundled with Flutter	Implement logic and data processing
VS Code	Latest stable	Development environment
Supabase	Stable self-hosted	Auth, database, file storage

PostgreSQL	Supported version	Store structured application data
Git	Latest stable	Version control

3.2.2 Hardware Specification

Hardware	Minimum Requirements
CPU	Dual-core 64-bit, 2.0 GHz or equivalent
RAM	4 GB
Drive Storage	10 GB available
Operating System	Windows 10+, Linux, or macOS
Additional Notes	Modern browser; internet/local network for backend access

4.0 Testing Methodology

4.1 Testing Model

- **(To be completed — e.g., unit, integration, system, UAT)**
- Focus areas:
  - Template design and customization
  - CSV import and data mapping
  - Individual and batch generation
  - PDF output consistency
  - Authentication and data persistence

4.2 Testing Tools

- **(To be completed — e.g., Flutter test, browser dev tools, Postman, etc.)**

5.0 Conclusion

- Proposed a **web-based certificate generator** using Flutter Web
- Addresses manual effort, inconsistency, and poor reuse
- Key features:
  - Layered design + reusable templates
  - Centralized recipient data management
  - Manual + CSV-based individual/batch generation
  - Consistent PDF output
- Methodology: **Throwaway Prototyping** to reduce technical risk
- Expected outcome: Simplified, organized, and reliable certificate preparation

Thank You

Questions & Discussion